Stereo Vision and Depth Estimation

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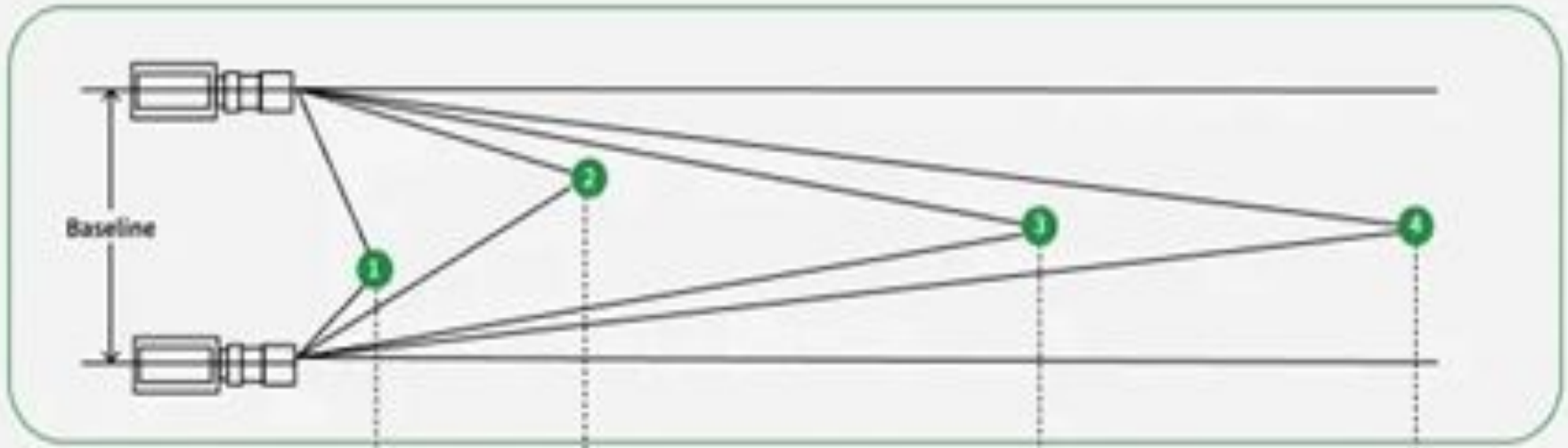
- Stereovision (or stereo vision) is a technique in computer vision that uses two or more cameras placed at different viewpoints to ***simulate human binocular vision.***
- It allows the perception of depth by identifying corresponding points in the images taken from each camera.
- The key idea is to estimate ***the disparity between two images to calculate depth***. It is widely used in robotics, autonomous vehicles and 3D reconstruction.



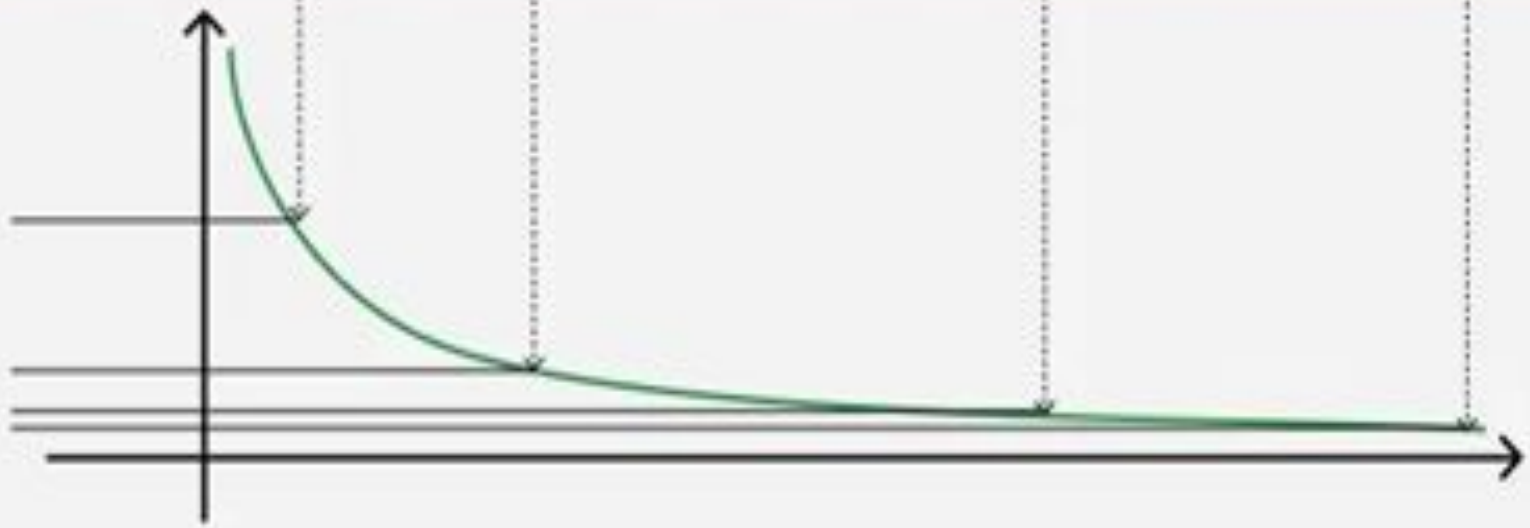
Key Features are:

- **Passive depth sensing**: Unlike LiDAR, it doesn't emit signals but uses ambient light.
- **Real-time capability** with optimized algorithms.
- **Low cost** compared to active depth sensors.
- **Dense depth maps** can be created from rectified images.
- **Non-intrusive** and suitable for indoor/outdoor environments.
- **Scalable** for multi-view stereo setups.
- **Accurate spatial mapping** for close and medium-range scenes.
- **Can be integrated** with other vision techniques.

Depth Estimation Setup



Disparity vs Distance in Depth Estimation

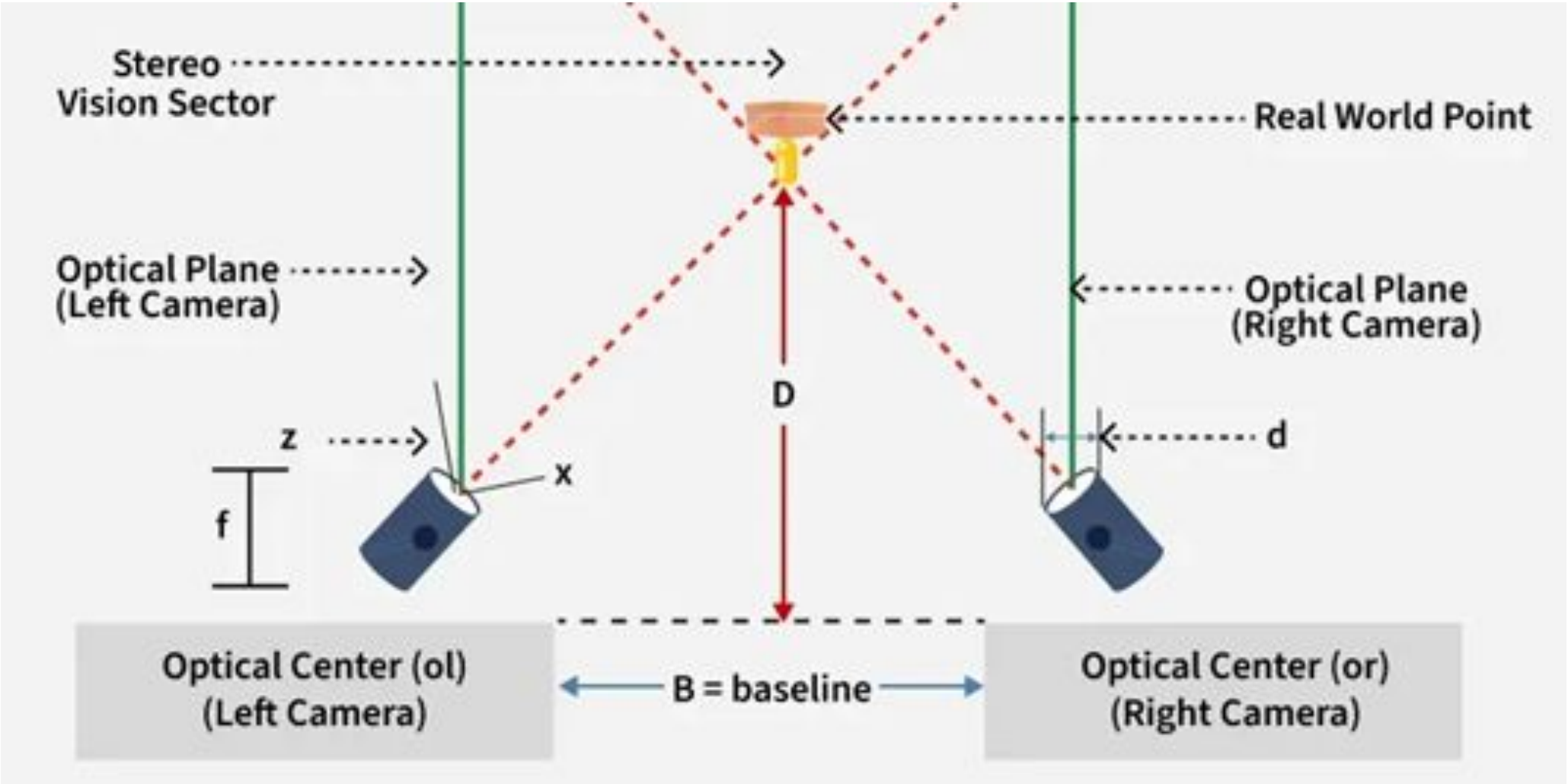


- **Stereo Camera Setup** : Two cameras with a known and fixed baseline.
- **Camera Calibration** : Intrinsic and extrinsic parameter estimation for accurate geometry.
- **Image Rectification** : Aligns image planes so epipolar lines are horizontal.
- **Feature Detection / Stereo Matching** : Detects or matches corresponding points (using methods like SGM or block matching).
- **Disparity Estimation** : Measures pixel shifts (disparity) between the two images.
- **Depth Computation** : Calculates depth using triangulation.
- **Depth Map Generation** : Produces a pixel-wise 2D array representing distances.
- **3D Reconstruction (Optional)** : Converts disparity and camera parameters into 3D coordinates or point clouds.

Mathematical Representation

- To calculate disparity and depth:
- Let:
- f – Focal Length in pixels
- B – Baseline (Distance between cameras)
- d – Disparity
- Depth $Z = (f \times B) / d$
- Disparity $d = x_{left} - x_{right}$

This formula assumes the images are rectified and the epipolar lines are horizontal.



Continuation…..

- Stereovision assumes that ***both cameras are calibrated and aligned***.
- A rectified stereo image pair enables easier disparity computation by aligning corresponding epipolar lines horizontally.
- By analyzing the slight differences (parallax) between the two images, the system can triangulate depth.
- It ***mimics the natural depth perception*** mechanism in human vision.

- **Capture Images**: Use two synchronized cameras with a fixed baseline to capture left and right images of a scene.
- **Camera Calibration**: Estimate **intrinsic parameters** (focal length, principal point) and **extrinsic parameters** (rotation and translation between cameras).
- **Image Rectification**: Warp both images so that epipolar lines are aligned horizontally. This simplifies correspondence search to one dimension (along scanlines).
- **Stereo Matching / Disparity Estimation**: Find corresponding pixels in both images using Block Matching, Semi-Global Matching (SGM), Graph Cuts and Deep Learning.
- **Disparity Map Construction**: Generate a disparity image where each pixel stores the disparity value (in pixels).
- **Depth Estimation (Triangulation)**: Compute depth using the formula: $Z = f \cdot Bd$ $Z = \frac{df \cdot B}{}$
- **Depth Map and Visualization**: Generate a 2D depth map or point cloud to represent 3D structure.

Techniques used in Stereo Vision

- Block Matching
- Semi-Global Matching
- Graph Cut Optimization
- Dynamic Programming
- Local and Global Stereo Matching Algorithms

Advantages of Stereo Vision

- No need for active sensors.
- Relatively low-cost with commercial stereo cameras.
- Useful in both indoor and outdoor environments.
- Good depth resolution for nearby objects.
- Can be used in real-time systems.

Applications of Stereo Vision

- Autonomous Vehicles for obstacle detection and navigation.
- Robotics for grasping and interaction with environments.
- Medical Imaging.
- Industrial Inspection